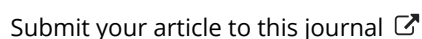
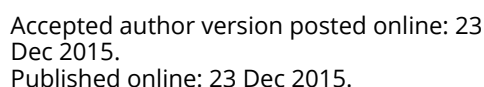


Javier A Menendez

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Metabolic control of cancer cell stemness: Lessons from iPS cells

Javier A Menendez^{1,2,*}

¹Metabolism & Cancer Group; Translational Research Laboratory; Catalan Institute of Oncology; Girona, Spain; ²Molecular Oncology Group; Girona Biomedical Research Institute; Girona, Spain

The Nobel prized discovery of nuclear reprogramming is swiftly providing mechanistic evidence of a role for metabolism in the generation of cancer stem cells (CSC). Traditionally, the metabolic demands of tumors have been viewed as drivers of the genetic programming detected in cancer tissues. Beyond the energetic requirements of specific cancer cell states, it is increasingly recognized that metabolism *per se* controls epi-transcriptional networks to dictate cancer cell fate, i.e., metabolism can define CSC. Here I review the CSC-related metabolic features found in induced pluripotent stem (iPS) cells to provide an easily understandable framework in which the infrastructure and functioning of cellular metabolism might control the efficiency and kinetics of reprogramming in the re-routing of non-CSC to CSC-like cellular states. I suggest exploring how metabolism-dependent regulation of epigenetics can play a role in directing CSC states beyond conventional energetic demands of stage-specific cancer cell states, opening a new dimension of cancer in which the “physiological state” of CSC might be governed not only by cell-autonomous cues but also by local micro-environmental and systemic metabolo-epigenetic interactions. Forthcoming studies should decipher how specific metabolites integrate and mediate the overlap between the CSC-intrinsic “micro-epigenetics” and the “upstream” local and systemic “macro-epigenetics,” thus paving the way for targeted epigenetic regulation of CSCs through metabolic modulation including “smart foods” or systemic “metabolic nichotherapies.”

mechanistically supports one of the most contemporary hypotheses in cancer pathophysiology, which states that the molecularly distinct subpopulation of so-called cancer stem cells (CSC) are responsible for many if not all aspects of tumorigenesis including the existence of deadly metastases, and also the clinical failure of the majority of available cancer therapies. The question of whether CSC ultimately originate *via* mutations that occur in normal stem cells, or from differentiated cells which reacquire stem cell attributes i.e., the acquisition of capacities to self-renew and to maintain multipotency or pluripotency through dedifferentiation, remains to be answered unequivocally.^{2–11} Nonetheless, the striking similarity of the molecular features shared between iPS cell generation and tumorigenesis is providing key mechanistic insights on how CSC could actually arise, in some cases, from differentiated cells through a process of “pathological nuclear reprogramming.”^{12–21}

A proof-of-concept demonstration of the close association between acquisition of stem cell properties by induced pluripotency and CSC-driven tumorigenesis has been recently carried out in a landmark study, showing that transient *in vivo* expression of reprogramming factors generates tumors with altered epigenetic states which cause abnormal growth of incompletely reprogrammed cells.²² Though these findings are the first to confirm that premature termination of induced pluripotency can result in cancer development, it should be noted that oncogenic-transformed cells and iPS cells generated from common parental fibroblasts have been found to represent highly related, yet distinct, cell types based on expression profiling,¹⁵ thus suggesting that they should

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*Correspondence to: Javier A Menendez; Email: jmenendez@iconcologia.net; jmenendez@idibgi.org

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The 2012 Nobel prized discovery of induced pluripotent stem (iPS) cells¹

share common “cellular ancestors” that develop along an equivalent molecular pathway(s) before they diverge. Indeed, a model comparing malignant transformation and (non-malignant) nuclear cell reprogramming demonstrated that differentiated cells should first acquire epigenetic changes that lead to a downregulation of the differentiation machinery, which is paralleled by an activation of glycolysis and other metabolic pathways.¹⁵ Crucially, only then are the oncogenic or the pluripotent phenotypes fully acquired, depending on other stimuli such as stemness factors. Moreover, whereas reprogrammed pluripotent stem cells can acquire oncogenic traits, the converse is not true because oncogenic cells cannot acquire the *bona fide* pluripotent state possessed by stem cells.¹⁵

If the acquisition of stem cell properties in induced pluripotency is closely associated with CSC-driven tumorigenesis, it then follows that determining the mechanisms that positively regulate the efficiency and kinetics of somatic reprogramming to iPS cellular states may provide a proof-of-concept validation for the novel self-renewing tumor-initiating mechanisms that regulate both the number and aberrant functionality of CSC.²³ Following this line of reasoning, Tung and

Knoepfler²⁴ have recently reviewed the shared epigenetic machinery by which pluripotency and oncogenicity are established and regulated. Interestingly, while the close similarity between iPS cell generation and the acquisition of CSC is shedding new light on the roles of *bona fide* oncogenes, tumor suppressor genes, transcription factors and chromatin regulators, in mediating the transition from differentiated-to-stem cell states in cancer tissues, an increasing number of experimental studies have consistently revealed that, similar to embryonic and adult stem cells, iPS cells are metabolically distinct from their differentiated counterparts.²⁵⁻³² Moreover, the precise metabolic properties of stem cells appear to be functionally relevant for stem cell identity and specification regardless of their cellular sizes or cell duplication dynamics, implicating a metabolism-centric regulation of stemness and cell fate.

Here I briefly review the CSC-related metabolic features found in iPS cells, to provide an easily understandable framework in which the infrastructure and functioning of cellular metabolism might operate as a key molecular constraint controlling the efficiency and kinetics of stemness reprogramming in the re-routing of non-CSC to CSC-like cellular states

during cancer genesis and progression (Fig. 1).

Metabolism and Cancer Stemness: Lessons from iPS Cells

First lesson: OXPHOS-to-glycolysis bioenergetic retuning

A shift in the balance between mitochondrial oxidative phosphorylation (OXPHOS) and glycolysis that reconfigures cellular anabolic requirements to those commonly found in the Warburgian bioenergetic transformation of cancer tissues (i.e., high glycolytic carbon flux and increased decoupling from ATP production in mitochondria) precedes the acquisition of stemness traits in iPS cells.³³⁻⁴⁷ Similar to well-recognized genetic and epigenetic factors, OXPHOS-to-glycolysis bioenergetic resetting appears to operate as a crucial enabling regulator of nuclear reprogramming because the self-renewal and pluripotency attributes cannot be efficiently acquired in the presence of an inadequate bioenergetic metabolotype. Thus, efficiency of reprogramming is greater the closer the glycolytic/OXPHOS energy metabolism profiles of the starting somatic cells are to those of embryonic stem cells (ESC). Moreover, bioenergetic resetting has an early and active role in reprogramming since manipulations that inhibit glycolysis reduce, whereas augmenting glycolysis enhances, reprogramming efficiency.⁴⁸ Therefore, only when a critical, very early step of engagement to the stemness metabolotype is correctly initiated can transcriptional regulators of self-renewal and pluripotency induce additional endogenous factors to acquire a *bona fide* stem cell state. This process implicates a hierarchical role of an adequate bioenergetic competence during nuclear reprogramming. Accordingly, induced pluripotency can be achieved with a combination of only one stemness transcription factor and small molecules able to facilitate the metabolic transition from OXPHOS to glycolysis.⁴⁸

Although the absence of the reprogramming factors c-Myc and Lin28, which are able to directly regulate metabolism by enhancing glycolysis and repressing mitochondrial respiration,⁴⁹ does not impede

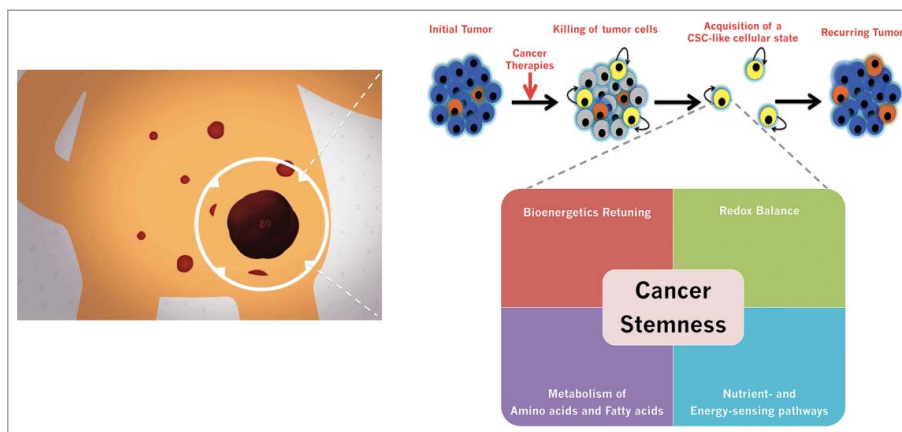


Figure 1. Metabolic restructuring and the acquisition of CSC cellular states: Beyond the nuclear-centric view of cancer stemness. The inherent aggressiveness of carcinomas appears to derive not from the pre-existing content of CSC, but rather from the intrinsic proclivity of a given tumor tissue to generate new CSC-like cellular state from non-CSC cell populations. We are accumulating evidence that enabling such cellular plasticity potential in cancer tissues requires an underexplored integration of metabolic stimuli with the epigenetic control of cell fate.^{122,135} Metabolic-related processes including bioenergetic retuning, redox balance, restructuring of biosynthetic requirements, and relative shifts in metabolic sensors and regulators are not secondary consequences of the acquisition of cancer cell stemness but they exert an important modulatory role.^{140,141}

metabolic resetting in iPS cells (i.e., metabolic resetting is a property of pluripotency which is independent of c-Myc and Lin28),³⁶ it should be noted that the OXPHOS/glycolysis signature of iPS cells is closely related to their intrinsic tumor-initiating capacity and differentiation potential. The inclusion of c-Myc, a key signaling node in cancer cell metabolism, significantly potentiates the glycolytic behavior and the tumorigenic incidence of derived iPS cells;⁵⁰ conversely, removal of c-Myc decreases the tumorigenicity of iPS cells and facilitates OXPHOS-dependent lineage commitment and terminal differentiation. Also, iPS cells drastically limit the activity and cellular content of the mitochondrial H⁺-ATPase synthase, a molecular feature that correlates directly with OXPHOS activity and inversely with the rate of glucose utilization by aerobic glycolysis in tumor tissues.⁵¹⁻⁵⁹ **Somatic cell reprogramming involves a dramatic downregulation of the catalytic β 1-F1-ATPase subunit and a significant increase in the expression of ATPase inhibitor factor 1 (IF1)**, a marker that is expressed in adult stem cells (ASC) for maintaining the activity of aerobic glycolysis, but is not expressed in equivalent differentiated cells.⁶⁰

Although little is known about the bioenergetic resetting of CSC, it appears that analogous to iPS cells, a direct link might exist between the occurrence of a metabolic switch from OXPHOS to aerobic glycolysis and the occurrence and maintenance of CSC cellular states. Compared to their more differentiated progeny, a stem cell cellular state might necessarily promote changes in the bioenergetic metabolism because CSC appear to preferentially perform glycolysis over OXPHOS, at least in some cancer types. Accordingly, recent studies have confirmed that a metabolic switch to glucose metabolism is a critical promotional event in the epithelial-to-mesenchymal (EMT)-driven CSC-like phenotype.^{61,62} **Epigenetic silencing of the gluconeogenic enzyme fructose-1,6-biphosphate, which catalyzes the energy-consuming conversion of fructose 1,6-biphosphate to fructose-6-phosphate, is employed by CSC as a mechanism of glucose flux maintenance via glycolysis** and other associated biosynthetic pathways.

Indeed, because its presence in the cell-culture milieu has been shown to significantly increase the percentage of CSC in the overall cancer cell population, **glucose appears to act as an essential nutrient for CSC**.⁶³ Conversely, glucose starvation is sufficient to cause a rapid depletion of the CSC subpopulation *in vitro*. A recent proteomic and targeted metabolomic analysis of the main differences between breast CSC and their differentiated counterparts has identified a metabolic phenotype associated with the stem-like condition, indicating that breast CSC shift from mitochondrial OXPHOS toward fermentative glycolysis.⁶⁴ Of note, whereas the treatment of fibroblasts undergoing nuclear reprogramming with 2-deoxyglucose (2DG), a general inhibitor of glycolysis, blunted the generation of iPS cells,³⁶ 2DG treatment inhibited breast CSC proliferation, thus revealing how the glycolytic requirement for iPS generation may be a potentially effective strategy to target breast CSCs.⁶⁴

Second lesson: Redox regulation

Beyond the activation of multiple, redundant mechanisms limiting energy production by OXPHOS in preference to glycolysis and its biosynthetic pathway branches (e.g., the pentose phosphate pathway), **the cellular redox state also impacts the balance between differentiated and stem cellular states**.⁶⁵ The oxidation-reduction state appears to parallel the stem cell fate since the accumulation of reactive oxygen species (ROS) has been **shown to be minimized in iPS cells via a reduction in substrate oxidation and respiratory coupling, which is accompanied by the activation of antioxidant stress genes**.^{38,66,67} In contrast, an oxidative metabolome, which effectively increases the overall cellular oxidation state during differentiation,^{43,68,69} appears to replace the unsaturated metabolome possessed by stem cellular states, including those of iPS cells.

Since strict regulation of specific redox species, such as the NAD⁺/NADH ratio, is directly impacted by glycolytic and mitochondrial activities that change during reprogramming, the NAD⁺/NADH redox state might have a key role in driving the stem cell fate.⁷⁰ Nicotinamide, a

precursor of NAD, significantly lowers the barrier to reprogramming to stemness by accelerating cell proliferation and protecting cells from apoptosis and senescence through alleviating oxidative stress, ROS accumulation, and subsequent mitochondrial potential collapse during iPS generation.⁷¹ Nicotinamide's ability to overcome pluripotency deficits and reprogramming barriers has yet to be evaluated in the acquisition and maintenance of CSC cellular states and future studies should establish whether an adequate NAD⁺ content is important for enhancing the resistance to stress in CSC. Accumulating evidence suggests that, in contrast to differentiated cancer cells where ROS levels are increased, CSC maintain low levels of ROS,⁷²⁻⁷⁵ and thus exhibit redox patterns similar to the corresponding normal stem cell. Accordingly, an increased reliance on glucose metabolism reduces the levels of ROS to promote EMT and CSC-like phenotypes.^{61,62} While the ratio of reduced glutathione (GSH), the primary intracellular antioxidant, to oxidized (GSSH) glutathione decreases, as does the level of NADH, during stem cell differentiation, CSC possess enhanced mechanisms of protection from stress induced by ROS that might render them resistant to chemo- and radiotherapy through an upregulation of GSH synthesis.⁷⁶⁻⁷⁸ Consequently, new strategies aimed to induce ROS *via* depletion of cellular glutathione can be viewed as promising therapeutic approaches against CSC.⁷⁹⁻⁸¹

Third lesson: Metabolism of amino acids and fatty acids

The Warburg effect is associated with a survival advantage as well as the generation of substrates such as nucleotides, amino acids, and fatty acids.⁸²⁻⁸⁵ In this regard, the stemness feature also takes advantage of the ability of the mitochondrial tricarboxylic acid (TCA) cycle intermediates to be siphoned off into amino acid and lipid biosynthesis, which have been shown to be required for self-renewal and differentiation of ES and iPS cells. In a series of studies employing combinatorial approaches of metabolomics, nutrition, and genetics, the amino acid threonine was identified as an essential

nutrient for mouse and human ESC.⁶⁹ Beyond its recognized role as a protein precursor, threonine dehydrogenase (TDH)-mediated catabolism of threonine provides glycine through one-carbon metabolism for biosynthesis of purines to support DNA replication and the epigenetic modifications required for self-renewal and maintenance of pluripotency.⁴² The high-flux metabolic state of iPS cells relies not only on a high dependence on threonine catabolism but also on large amounts of methionine; thus, iPS cells display regulatory systems to maintain a constant level of intracellular methionine and S-adenosylmethionine (SAM), a key regulator for maintaining undifferentiated iPS cells and regulating their differentiation.⁸⁶ Whether the acquisition of CSC cellular states and CSC self-renewal and differentiation similarly relies on TDH-related purine biosynthesis and/or methionine metabolism remains an unexplored area in the field of CSC biology. Nevertheless, the fact that threonine provides a substantial fraction of both cellular glycine and the acetyl-coenzyme A required for SAM synthesis, together with the recently recognized ability of SAM to influence trimethylation of histone H3 lysine 4 (H3K4me3),⁸⁷ provides a probable epigenetic mechanism by which modulation of a metabolic pathway might directly influence aberrant stemness in cancer tissues.⁸⁸⁻⁹⁰

When the catalytic activities of acetyl-CoA carboxylase (ACACA) and fatty acid synthase (FASN) lipogenic enzymes are inhibited, the efficiency of reprogramming is significantly decreased. Coincidentally, ACACA and FASN are highly expressed in iPS cells.⁵¹ Lipids that participate in signaling cascades, such as arachidonic acid, diacylglycerol, and prostaglandins, are also among the most predominant metabolites found in iPS cells.⁴² It is known that higher expression levels of lipogenic genes and proteins such as FASN are found in CSC subpopulations of breast cancer cell lines, and that upregulation of *de novo* fatty acid biogenesis is a prerequisite for the formation of premalignant lesions by endowing CSC survival.^{91,92} Beyond the fact that fatty acids can play a shared role in normal and cancerous stem cell energy generation *via*

fatty acid oxidation, pharmacological inhibition of ACACA and FASN efficiently impedes the formation of mammospheres in a fatty acid-dependent manner, strongly suggesting that the self-renewal and survival of CSC can be directly impacted by *de novo* lipogenesis, lipid metabolites, and lipid catabolism.^{93,94} With regard to the latter, overexpression of monoacylglycerol lipase (MAGL) in nonaggressive cancer cells is sufficient to increase their pathogenicity by recapitulating a fatty acid network enriched in oncogenic signaling lipids that promote migration, invasion, survival, and *in vivo* tumor growth.^{95,96} Given the unique role of MAGL in providing lipolytic sources of free fatty acids for the synthesis of oncogenic signaling lipids, CSCs might co-opt lipolytic enzymes such as MAGL to translate their lipogenic state of stem cells into an array of protumorigenic signals. The recent discovery that cancer adaptation to antiangiogenic treatments, which generate hypoxic and nutrient-starved tumor microenvironments that suppress cell cycle progression but enrich non-proliferating CSC,⁹⁷⁻¹⁰³ involves a significant upregulation of lipid synthesis that fuels tumor regrowth and metastasis after angiogenic therapy withdrawal,¹⁰⁴ strongly supports the notion that key enzymes involved in lipid metabolism such as FASN could play key roles in the reprogramming of CSC cellular states.

Fourth lesson: Nutrient- and energy-sensing pathways

By integrating many metabolic signals, the mammalian target of rapamycin (mTOR)/AMP-activated protein kinase (AMPK) signaling network appears to operate as a critical “metabolic rheostat” that can direct the generation and maintenance of stem cells.^{26,105} A timely and precise regulation of the activation/deactivation status of mTOR, a kinase with a central role in sensing O₂, nutrients (glucose, amino acids), and growth factors, critically determines the successful reprogramming of somatic cells to iPS cells.¹⁰⁶⁻¹⁰⁹ Accordingly, an early short burst of mTOR suppression followed by restoration of mTOR activity at a later stage is required for successful reprogramming. The stemness and oncogenic

transcription factor, *Sox2*, exclusively initiates the resetting of the metabolic infrastructure by suppressing mTOR activity in a timely manner before the acquisition of stemness. Continuous mTOR signaling would evoke active cycling and exhaustion of normal stem cells or fail-safe mechanisms (e.g., cellular senescence, apoptosis, or terminal differentiation) that may extinguish the evolution of pre-malignant stem cell clones or prevent efficient reprogramming of somatic cells. Conversely, the cells can exploit the proto-oncogenic and reprogramming signaling emanating from active mTOR only by evading the above-mentioned cancer-protecting systems, which is an escape mechanism that depends not only on the genetic make-up of the cells (e.g., loss of tumor suppressor genes) but also on the activation status of energy-sensing regulators and mTOR suppressors, such as AMPK. Consequently, iPS cells can avoid the inhibition of stemness-related anabolic pathways by downregulating the catalytic activity of AMPK,³³ a metabolic master switch that senses and decodes intracellular changes in the energy status and that, upon activation, can “switch off” biosynthetic pathways by negatively regulating the Warburg effect. AMPK activation imposes a metabolic flow away from the required pro-immortalizing glycolysis that fuels the induction of stemness and pluripotency, endowing somatic cells with an energetic infrastructure that is protected against reprogramming.^{110,111} Thus, AMPK activation establishes a metabolic barrier to reprogramming that cannot be bypassed even by a deficiency of key tumor-suppressor genes such as p53, which is a key element that greatly improves the efficiency of stem cell production.

Regarding cross talk between mTOR and *Sox2*, which regulates early genetic reprogramming and the acquisition of stemness not only in iPS cells but also CSCs, the metabolic barrier evoked by AMPK activation impacts the genetic regulation of normal and cancerous stem cell states by regulating core transcriptional regulators. AMPK activation impedes early reprogramming by preventing transcriptional activation of *Oct4*, the master regulator of the pluripotent state.¹¹⁰⁻¹¹²

Moreover, AMPK activation facilitates the specific elimination of *Oct4*-positive teratoma-initiating pluripotent stem cells that are intermixed with non-tumorigenic iPS cell derivatives, strongly suggesting that an adequate functioning of the metabolic infrastructure might be an indispensable component of the CSC machinery.¹¹¹ Using an *in vitro* model of *de novo* generation of CSC-like states through nuclear reprogramming in an established breast cancer cell line, we have recently shown that the transcriptional suppression of mTOR repressors (i.e., *PRKAA1*, which codes for the catalytic $\alpha 1$ subunit of AMPK; *DDIT4/REDD1*, a stress response gene that operates as a negative regulator of mTOR; and *DEPTOR*, a naturally occurring endogenous inhibitor of mTOR activity) appears to be an intrinsic process that occurs during the acquisition of CSC-like properties by differentiated breast cancer cells.¹⁷ AMPK and mTOR alone, or through their substrates, operate as interconnected metabolic controllers that fine-tune the stemness activity of core reprogramming factors such as *Sox2* and *Oct4*, rather than acting as *on/off* reprogramming-to-stemness switches. It is therefore not surprising that the potential oncology applications for biguanides,¹¹³ which can activate AMPK leading to inactivation of mTOR, appear to be closely related to their differential metabolic effects during the inducible generation of CSC.¹¹⁴⁻¹¹⁷

Cancer Metabostemness: A Waddingtonian Perspective of the Metabolic Control of Stemness and Cell Fate in Cancer Tissues

An intuitive perspective of the metabolic control of stemness in cancer tissues can be offered using Waddingtonian landscapes, which accurately model the complex network of molecular barriers governing cell fate transitions. In 1957, Conrad Waddington proposed a metaphorical representation of cell differentiation phenomena in which pluripotent stem cells (or, in the analogy, “balls”) are positioned at the top of a hill, progressively losing potential while journeying

downhill into different valleys representing irreversible, fully differentiated cellular states.¹¹⁸ Thus, once the group of balls at the top start to roll downhill and select a route influenced by various genetic and environmental factors, the balls are committed (i.e., they become differentiated cells) and ultimately land at the bottom. Movement upwards will be difficult, inferring therefore that commitment of cells to specific tissues and organs is fundamentally one directional. They cannot roll back uphill to the bifurcation and they cannot climb over the portion of land between the valleys because reversal of the commitment process will trigger undesirable damaging effects (e.g., cancer).^{119,120} In this scenario, the radical change of identity required for cellular reprogramming of non-CSC to CSC states is expected to be not easily accomplished because by the time all the organs are formed most cells are highly committed to their specific tissue and, in many cases, few ASC remain. However, although the one-way process of commitment generally suppresses cancer until late in life, the groundbreaking discovery of Yamanaka and colleagues demonstrating that terminally differentiated cells can be pushed “upwards” to an original pluripotent stem cell state,¹ is rapidly and revolutionarily providing strong mechanistic evidence for the convergence and commonality of iPS generation and CSC states. Genetically and microenvironmentally induced pathological reprogramming, by operating as a sideward deviation of the self-organizing property of developmental epigenetic landscapes, radically modifies our current perception of cancer genesis and tumor behavior.

Cancer is beginning to be understood as a disease of cellular reprogramming (or a disease affecting cellular differentiation) in which many driving forces such as the loss of tumoral suppressors or the aberrant activation of certain transcription factors, signaling cascades, or epigenetic regulators, appear to share a permissive role which alleviates the developmentally unfavorable process of acquiring tumor-initiating and/or metastasis-initiating capabilities possessed by CSC.^{13,20-23} Because the notion that reprogramming can occur through non-strictly genetic

cues (i.e., “master stemness genes” can be dispensable for cell fate reprogramming) has been substantiated through the observation that stemness in somatic cells can be achieved using a chemical cocktail that exclusively regulates the signaling pathways and levels of metabolic intermediates,¹²¹ we recently coined the term “metabostemness” to refer to the cellular metabolotype or metabolic parameters (e.g., OXPHOS-to-glycolysis resetting of bioenergetic metabolites, metabolic cofactors of epigenetic modifiers, *bona fide* oncometabolites, etc.) that might causally control or, more importantly, functionally substitute the epitranscriptional orchestration of genetic reprogramming that redirects normal and non-CSC tumor cells toward less-differentiated CSC cellular states.¹²² From a mathematical perspective, a given cellular metabolotype might not only dictate the plasticity of the original cell that is targeted to be reprogrammed by early or late genetic and/or epigenetic hits but also, by removing, diminishing, or modifying the nature of “molecular barriers” present in the Waddington epigenetic landscapes, it can operate as a *bona fide* accelerator of the reprogramming processes, thus allowing not only ASC but also (normal, premalignant or tumor) differentiated cells to more easily and rapidly enter, or re-enter, into CSC cellular macrostates (Fig. 2).

Metabolo-epigenetic reprogramming of CSC functions

In cancers with a stem cell origin, such as haematopoietic malignancies, certain cellular metabolotypes could sculpt the landscape in a manner that leads to previously normal stem cells or early progenitors becoming stuck in close proximity or in the same state space attractor of immature stem-like regions of the landscape,¹²³ thus increasing the number of undifferentiated cells that may be targetable by oncogenic mutations. Indeed, an invaluable model supporting the notion that certain metabolotypic alterations might operate as pivotal molecular events rendering stem cells susceptible to the metabolic rewiring necessary for the acquisition of aberrant stemness and, concurrently, of refractoriness not only to apoptosis but also to differentiation, arises from cancers with a stem cell origin in which gain-of-function

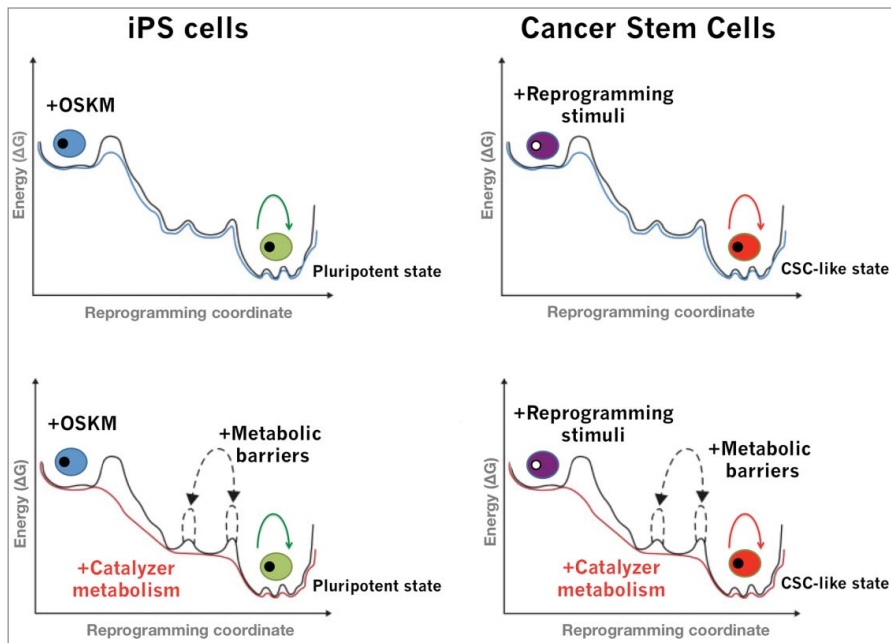


Figure 2. Metabolic-driven control of cellular reprogramming: An integrated view of cell fate transitions. Reversible epigenetic barriers that can be overcome given the correct stimuli preserve cellular fate. Thus, once a certain combination of epigenetic changes has been acquired, cells can assume a new identity (iPS cells or CSC-cells). At a cellular level, both induced pluripotency (*left panels*) and acquired cancer stemness (*right panels*) should be viewed as multi-step processes that result in a change of cell identity or differentiation potential where nascent iPS cells and non-CSC cells should face the same epigenetic barriers to alter cell identity.¹⁴² Moreover, the “end product” is in both cases an immortal cell with tumor-initiating capacity. Figure illustrates the “energy landscape” experienced by cells under reprogramming conditions under different epigenetic perturbations (modified from ref. 143). Energy peaks represent barriers in the reprogramming path, where higher barriers correspond with low conversion rates. The black line represents the energy plot of the path from somatic cells to either iPS cells or CSC-like states in response to reprogramming stimuli (e.g., pluripotency-promoting transcription factors, canonically Oct4, Sox2, Klf4, and Myc [abbreviated as OSKM]). Certain cell metabolotypes are more susceptible to *de novo* reprogramming indicating a more permissive epigenetic environment. Certain metabolic shifts may represent an early *bona fide* reprogramming event (blue line) and, by having direct effects at the epigenetic level, certain metabolic features operate as catalyzers that reduce energy barriers and accelerate reprogramming (red line). Conversely, imposed barriers (dashed black lines) can occur via certain metabolic conditions that might inhibit and impair the epigenetic rewiring during reprogramming (e.g., AMPK activation hampers the reactivation of the stemness factor Oct4).^{110–112} Indeed, CSC are not irreversibly locked in a tumorigenic state but instead amenable to metabolite-epigenetic reversion into a phenotypically non-CSC state.

mutations in isocitrate dehydrogenase (IDH) generates the oncometabolite 2-hydroxyglutarate (2HG).^{124–130} Because IDH enzymes produce α -ketoglutarate, a cofactor for the TET family of DNA demethylases, neomorphic mutations of IDH that drive the aberrant neosynthesis of the TET inhibitor 2HG can be expected to mostly restrict the “methylation plasticity” that is required for the hierarchical transitions between stem cells and their differentiated progeny. 2HG-induced global hypermethylation prevents the demethylation of genes that are implicated in differentiation,

promoting a metabolically-driven increase in the number of stem cells that might occur prior to oncogenic mutations promoting proliferation.²¹

In the absence of IDH mutations, however, the accumulation of 2HG has been shown also to be part of the c-Myc-driven metabolic reprogramming phenomenon observed in biologically aggressive breast carcinomas that exhibit globally increased DNA methylation.¹³¹ The hypermethylation phenotype of 2HG-overexpressing breast carcinomas is characterized by a strong enrichment of a stem cell-like transcriptional signature, a

molecular feature that has been similarly observed in other solid tumors with non-stem cell origins such as intrahepatic cholangiocarcinoma, a deadly liver malignancy in which highly prevalent 2HG-producing IDH mutations subvert the hepatocyte differentiation/quiescence program to create a persistent pre-neoplastic state which is primed for transformation into adenocarcinoma by additional oncogenic mutations.¹³² Given the inherent cell attractor nature of nuclear reprogramming phenomena,^{119,120,123,133,134} we recently provided an intuitive perspective on how the commonly forgotten ability of metabolism to reshape the topography of Waddingtonian epigenetic landscapes can influence the probability that pathologically reprogrammed cells can stay trapped as developmentally immature, CSC-like states in the “high mountain” regions of the landscape.¹³⁵ We proposed that refining the cancer attractor hypothesis by merging “pathological nuclear reprogramming” with the specific metabolic remodeling traits of iPS cells offers a new integrative-systems perspective of the stem cell theory of cancer that explains why oncometabolic-like traits (i.e., the specific ways that CSC generate and use metabolic pathways and intermediate metabolites) converge to “encode” an immature, stem-like program in cancer tissue, regardless of the stem cell/non-stem cell source.

For cancers with non-stem cell origins, certain metabolotypes can permissively alleviate the “uphill,” unfavourable developmental process of “jumping back” from non-CSC differentiated valleys to high-altitude CSC attractors while concomitantly promoting the ground-state character of self-maintaining CSC-like states. Because CSC states can be viewed as inherently inevitable epigenetic deviations of Waddington’s landscapes, in which metabolism modifies the probability that not only stem cells/early progenitors but also normal or cancer-differentiated cells can find either pre-existing or *de novo* occupied, self-organizing attractors encoding dynamically robust CSC signatures, it would be interesting to determine whether the very strict cellular metabolotypes that are compatible exclusively with iPS cell generation truly represent a non-genetic dimension of cancer that generates *bona*

fide “aberrant attractors” (i.e., dead-end side valleys) that are necessarily located close to or in the same location of the developmentally immature, stem-like cellular states in these “higher mountains.” These aberrant attractors might prevent pathologically reprogrammed undifferentiated cells in cancer tissues from completing their physiological, predestined journey down to the differentiated cell type attractors.

Metabolism-dependent regulation of epigenetics^{122,135} plays a key role in directing cancer cell fate, including CSC. The two primary epigenetic codes, DNA methylation and histone modification, and the consequent epigenetic regulation of differentiation genes might be viewed as the pivotal molecular events that successfully integrate the recognized ability of cellular metabolism to competitively inhibit epigenetic regulation of cell differentiation with the process in which the stemness regulatory circuitry is established during nuclear reprogramming. Certain metabolic events and elite metabolites (e.g., the Warburg effect, oncometabolites such as 2-HG, sub-cellular compartmentalization of critical metabolic cofactors of epigenetic enzymes, etc.) might markedly lower the “energy barriers” separating non-CSC and CSC attractors, diminish

the average time of reprogramming, and increase the size of the basin of attraction of the macrostate occupied by CSC, thus promoting the ground-state character of the self-maintaining CSC cell state. Indeed, from a Waddingtonian perspective, the underexplored topic of differentiation therapy can be viewed in terms of metabolic interventions able to knock CSC cellular states “trapped” in attractors back to the physiological trajectories that lead to non-malignant, more differentiated cells. Although these speculations appear plausible, considerably more direct experimental support will be required before they can be accepted.

Metabostemness: From CSC energetics to micro- and macro-epigenetics (a corollary)

At one time, the cancer genomics era pushed Warburg’s metabolic cancer hypothesis into obscurity. The Nobel prized discovery of iPS cells, however, is rapidly delivering mechanistic evidence on how the metabolic control of stemness might causally participate in the generation of CSC cellular states.^{105,122,135} I here postulate that incorporating certain metabolic observations of iPS cell behavior into pathological nuclear reprogramming phenomena might provide an

integrative view of the stem cell theory of cancer that may radically amend the molecular understanding and clinical management of cancer diseases. Looking forward, we will continue to make advances in this area by exploring at least 2 provocative questions:

First, a systematic delineation of CSC-specific expression of metabolic enzymes, metabolites, and certain bioenergetic processes (e.g., Warburgian metabolism versus mitochondrial respiration) should definitively clarify whether the metabolic state rather than phenotypic marker expression is a crucial operational criterion that defines CSC. In this regard, advances in functional metabolomics should reveal how catabolic processes such as autophagy^{26,136,137} contribute to the maintenance of substrate and metabolite levels that support not only the bioenergetic and biosynthetic needs of CSC, but also the aberrant reprogramming of the epigenome in CSC.

Second, although we are only just beginning to understand how cellular metabolism can maintain the hallmark trait of self-renewal in CSCs, it is likely that metabolism has an indispensable role to directly influence the most relevant epigenetic alterations, from DNA methylation to chromatin organization, which

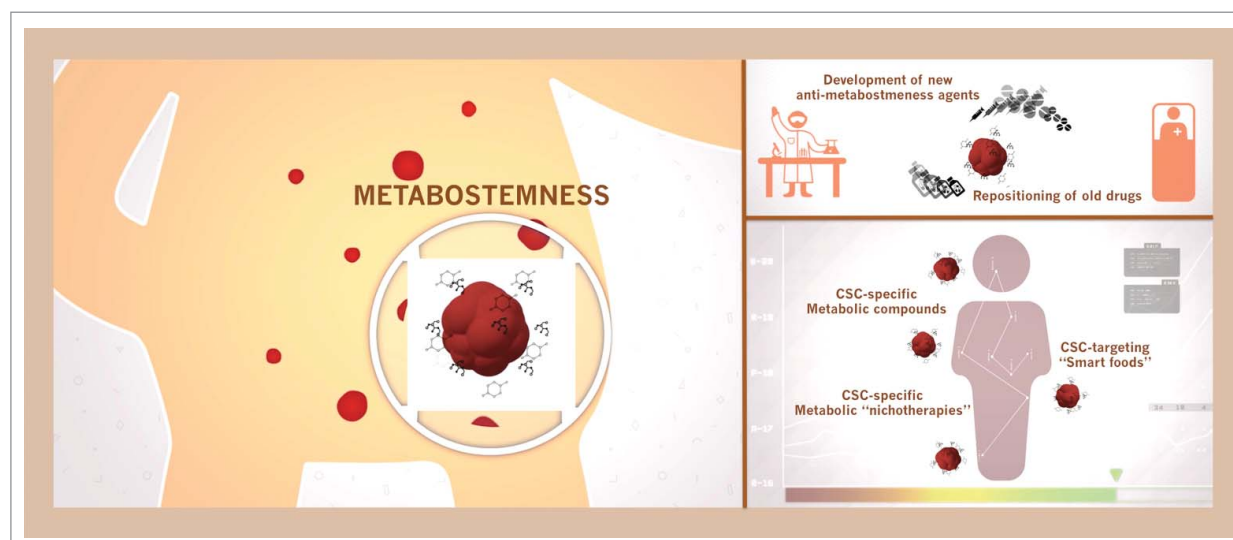


Figure 3. Metabostemness: A new therapeutic opportunity in cancer. Further knowledge is warranted to decipher the metabolic-driven control of cancer cell stemness in order to either improve novel discovery technologies or accelerate scanning of existing pharmacopoeia for repositioning candidates as a new therapeutic opportunity in cancer. New anti-metabostemness strategies should involve not only a conventional development of drugs directly targeting CSC metabolism,¹⁴⁴⁻¹⁴⁶ but also should consider the metabolic demands associated with the acquisition of CSC-like cellular states and therefore the influence of environmental metabolic factors and their usage in the epigenetic control of cell fate in cancer tissues.

contributes to the regulation of CSC features in tumor progression, metastasis and response to therapies. If we adopt the view that metabolism-dependent regulation of epigenetics plays a role in directing CSC states beyond matching energetic demands of stage-specific cancer cell states, we will be entering an unforeseen dimension of cancer diseases in which the “physiological state” of CSC is governed not only by cell-autonomous cues, but also by local micro-environmental and systemic metabolo-epigenetic interactions.

By deciphering how specific metabolites integrate and mediate the overlap between the CSC-intrinsic “micro-epigenetics” and the “upstream” local and systemic “macro-epigenetics,” as recently proposed by Miguel Ramalho-Santos,¹³⁸ we will pave the way for future developments to target epigenetic regulation of CSC through metabolic modulation including “smart foods” or systemic “metabolic nichotherapies” (Fig. 3).¹³⁹ The incorporation of bioenergetics and metabolo-epigenetics to the development of therapeutics targeting CSC should offer a new perspective on the frequently forgotten relevance of basic metabolic research in the post-genomic era of cancer research.

Disclosure of Potential Conflicts of Interest

No potential conflicts of interest were disclosed.

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